Project Title – TalentConnect – An Intelligent Student Collaboration & Academic Support Platform

Abstract:

TalentConnect is an intelligent, student-centered platform designed to facilitate collaborative learning, peer-to-peer mentorship, and task-based academic support within a secure digital environment. The system allows students to showcase their skills, request learning sessions, and offer academic assistance to peers in a structured and transparent manner. Integrated AI modules enhance the platform by enabling smart skill matching, personalized learning recommendations, skill verification, fraud detection, and an AI-powered chatbot for guidance. Through automated session scheduling, feedback mechanisms, and a secure payment gateway, TalentConnect ensures reliability, accountability, and accessibility. By combining peer learning with task support services, the platform promotes affordable education, mutual growth, and a technology-driven academic ecosystem that empowers students to collaborate, learn, and excel effectively.

Problem Definition:

In the current academic environment, many students face challenges in accessing affordable and reliable learning resources. High costs of online courses or private tutoring prevent some students from developing their skills effectively. Simultaneously, students often struggle to complete academic tasks due to time constraints, health issues, emergencies, or gaps in subject understanding.

Existing platforms either focus on paid tutoring or freelance task completion but lack a structured, student-centric system that enables peer-to-peer learning and task support in a secure and organized way. Moreover, there is no platform that leverages intelligent AI features to:

- Match learners with the most suitable mentors based on skills.
- Recommend complementary skills for personal growth.
- Detect misuse, cancellations, or fraudulent behavior.
- Verify claimed skill levels for credibility.
- Provide guidance through an AI-driven chatbot.

Therefore, there is a critical need for a centralized, technology-driven platform that facilitates collaborative learning, skill exchange, and academic assistance while ensuring trust, transparency, and accountability among students.

Core Functional Features

Feature 1: Skill Exchange & Rating System

- Students list their skills (e.g., "I know Python", "I can do Video Editing").
- Other students can request learning sessions.
- System schedules sessions by date & time and generates a meeting link (Google Meet integration).
- After the session:
    - Learner can give **ratings**
    - Learner can write a **review**

Feature 2: Paid Task Support Marketplace

- Students can upload academic tasks and set a price.
- Other students can accept and complete tasks.
- Payment is transferred securely to the task-completing student.
- After task completion, feedback ensures accountability and quality assurance.

Feature 3: System Controller

The **System Controller** acts as the central administrative hub of TalentConnect, providing oversight and management of all platform activities to ensure security, transparency, and smooth operation.

Key Functionalities:

1. **User Management**
    - Admin can **view all registered users** along with their profiles, skills, and activity history.
    - **Ban/Unban Users:** Restrict access for users violating rules or exhibiting fraudulent behavior; restore access when appropriate.
    - Monitor user interactions, feedback, and session histories to maintain trust.

2. **Task and Marketplace Oversight**
    - Track all tasks uploaded by students and their completion status.
    - Approve or flag suspicious or inappropriate tasks.
3. **Session and Skill Monitoring**
    - Oversee scheduled learning sessions and AI skill verification results.
    - Review AI-generated alerts for repeated cancellations, misuse, or suspicious behavior.
4. **Feedback & Rating Review**
    - Analyze session and task feedback for quality control.
    - Identify recurring issues and take corrective actions.

Benefits:

- Ensures **platform security** and misuse prevention.
- Maintains **trust and accountability** among students.
- Provides **actionable insights** for administrators to improve system operations.

AI-Enhanced Features

1️⃣ Smart Skill Matching

- The system intelligently filters available mentors based on the learner's requested skill.
- Example: If a student wants to learn **Flutter**, only users proficient in Flutter will appear as available mentors.
- Implementation: Cosine similarity algorithm + skill-tag database matching.

2️⃣ AI Skill Recommendation

- The platform suggests complementary or next-level skills for learners based on their current expertise.
- Example: A student listing **JavaScript** may receive a recommendation to learn **Node.js** for backend development.
- Implementation: AI recommendation engine using clustering and skill co-occurrence analysis.

3️⃣ AI Fraud Detection & Session Monitoring

- The system monitors user behavior to detect misuse, such as repeatedly canceling sessions.

- Users flagged for suspicious activity may receive automated warnings or negative impact on rating.
- Implementation: Behavioral analytics + anomaly detection model.

4️⃣ **AI Skill Verification**

- When a user claims proficiency (e.g., "I am advanced in Python"), the AI generates a small automated test to validate their skill level.
- Based on the evaluation:
  - **Pass → Skill approved**
  - **Fail → Skill rejected**
- Implementation: Auto-generated quizzes, coding problem evaluation using Python evaluation scripts.

5️⃣ **AI Chatbot & Learning Guidance**

- Interactive AI assistant for learners to ask skill-related queries.
- Provides guidance, learning paths, and resource suggestions.
- Example: "How do I start learning DSA?" → Chatbot provides stepwise roadmap and practice resources.
- Implementation: NLP-based AI chatbot integrated into the platform.

Tools and Technologies

Frontend Technologies

- **HTML5** – Provides the structure of web pages.
- **CSS3 / Bootstrap / Tailwind CSS** – Enables responsive and visually appealing UI design across devices.
- **JavaScript (ES6+)** – Adds client-side interactivity and dynamic behavior.
- **AJAX / Fetch API** – Supports asynchronous communication with the backend without page reloads.

Backend Technologies

- **Python (Flask / Django Framework)** – Handles server-side logic and request processing.
- **RESTful API Architecture** – Enables structured client-server communication and modular system integration.

- **JWT Authentication** – Provides secure login, session management, and role-based access control.

Database

- **PostgreSQL** – Primary relational database for storing user profiles, tasks, session schedules, feedback, and AI analytics.

Payment Integration

- **Razorpay API** – Secure payment gateway for task marketplace transactions.
- **Webhook Handling** – Automatically verifies payments and logs transaction details.

AI & Machine Learning Modules

- **Python Libraries:** NumPy, Pandas, Scikit-learn – For data processing, recommendation systems, and AI computations.
- **Smart Skill Matching:** Cosine Similarity algorithm matches learners with suitable mentors based on skills.
- **AI Skill Recommendation:** Clustering and co-occurrence analysis to suggest complementary skills for users.
- **AI Fraud Detection:** Behavioral analytics and anomaly detection to flag repeated cancellations or misuse.
- **AI Skill Verification:** Auto-generated quizzes and coding evaluations to validate claimed skill levels.
- **AI Chatbot:** NLP-based conversational assistant for providing learning guidance, roadmaps, and resources.